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SUPERSEDING
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MILITARY SPECIFICATION

FLOODLIGHT SET, ELECTRIC, DIESEL-ENGINE-DRIVEN, TRAILER MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a self-contained, trailer-mounted, floodlight set, equipped with a diesel-engine-driven, 6 kilowatt generator and a mechanically operated telescoping floodlight tower.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery.

|Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, 621 Pleasant Valley Road, Port Hueneme, CA 93043-4300, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.|

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MILITARY

MIL-C-104 - Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronics, and Associated Equipment).
MIL-C-3774 - Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-G-28554 - Generator Sets, Mobile Electric Power and Supplemental Equipment; Packaging of.
MIL-V-62038 - Vehicles, Wheeled: Preparation for Shipment and Storage of.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MIL-STD-1791 - Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which are current on the date of the solicitation (see 6.2).

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ASTM

ASTM D 3953 - Strapping, Flat Steel and Seals.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA AB 1 - Molded Case Circuit Breakers and Molded Case Switches.
NEMA WD 1 - General Requirements for Wiring Devices.
NEMA WD 5 - Wiring Devices, Specific Purpose.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 1 Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J560B - Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
SAE J1292 - Automobile, Truck, Truck-Tractor, Trailer, and Motor Coach Wiring.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, OH 44321.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 486 - Wire Connectors and Soldering Lugs for Use With Copper Conductors.
UL 498 - Attachment Plugs and Receptacles.
UL 1077 - Supplementary Protectors for Use in Electrical Equipment.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062-2096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The floodlight set shall consist basically of a trailer, mounted with a diesel-engine-driven generator, telescoping tower, with four 1,000 watt (W) metal-halide floodlights and a portable mast for mounting two 500W tungsten halogen floodlights. The floodlights and all electrical equipment shall be provided with a weatherproof storage compartment.

3.1.1 Dimensions and weight. The overall dimensions and weight of the floodlight set in towable condition, attachments retracted and stowed, shall not exceed 15 feet in length, 6.5 feet in width, 6.5 feet in height, and 2,500 pounds (lb) in weight.

3.1.2 Environmental and storage conditions. All components of the floodlight set shall withstand, without damage or impairment of operation, various environmental conditions including the following:

a. Rated performance, without additional auxiliary equipment, in any ambient temperature from -25 degrees Fahrenheit (F) to +125 degrees F (-31.7 degrees Celsius (C) to +51.7 degrees C), or when exposed to rainfall at a rate up to 3 inches per hour, impinging on the set at any angle from the vertical to 45 degrees from the vertical.

b. The set shall not be damaged by exposed storage at temperatures from -65 degrees F to 155 degrees F (-53.9 degrees C to +68.3 degrees C) at any possible relative humidity within this temperature range.

3.2 First article. The manufacturer shall furnish one floodlight trailer for first article inspection, tests and approval (see 4.3 and 6.2).

3.3 Standard product. Except where modified herein, it is intended that the equipment and its component parts shall be a regular commercial product of the manufacturer or his suppliers. All equipment and parts shall be new and unused.

3.4 Alternator. The alternating current alternator shall be a brushless design with a rated output between 1,600 and 2,000 revolutions per minute (rpm), of not less than 6,000W, 120/240 volts (V), 60 Hertz, single phase, alternating current. The alternator shall be rigidly coupled to the engine, permanently aligned and of single bearing design. The alternator shall have a brushless exciter and voltage regulated for ±10 percent and -8 percent maximum from no load through maximum rated load.

3.4.1 Power distribution equipment. All electrical components shall conform to applicable UL standards (including UL 486, UL 498, UL 1077) and to NFPA 70. The installation of all electrical components and systems shall conform to NFPA 70. The power distribution equipment shall consist of the power distribution panel, ballasts for the metal halide lamps, switches, receptacles, cord connectors and plugs, portable power cords, interconnecting wiring, fittings, and accessories, installed and connected as required to provide a complete, operable system.

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3.4.1.1 Power distribution panel. The power distribution panel shall be a circuit breaker panel mounted in a readily accessible location within the floodlight set housing. Sufficient circuit breakers shall be contained in the panel to provide independent control and circuit protection as defined in NFPA 70.

3.4.1.2 Circuit breakers. The circuit breakers shall conform to the requirements of NEMA AB 1, and UL 1077. All circuit breakers shall be approved for use as switches and shall be marked as such. Circuit breakers shall be sized to have trip ratings as required for protection against overloads, short circuits, and grounding in accordance with NFPA 70.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials (see 6.5.3) to the maximum extent possible without jeopardizing the intended use. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.6 Floodlight set components.

3.6.1 Generator set. The generator set shall be a unitized engine driven alternator. The generator set shall provide adequate electrical power for operation of the floodlight set at rated maximum load during environmental and test conditions specified herein. The generator set shall be mounted to minimize vibration.

3.6.2 Engine. The engine shall be a four-cycle, air- or water-cooled diesel which develops, at between 1,600 and 2,000 rpm, the necessary horsepower to meet the requirements specified for the floodlight set. The engine shall be furnished, as a minimum, with the following:

a. 12V electric starting system.
b. Alternator 12V, direct current, 2 to 5 ampere (A), adjustable trickle charger.
c. "Glow plug" preheater system.
d. Totalizing engine hour meter (control panel mounted).
e. Fuel tank level gauge.
f. Direct current voltmeter to indicate battery potential (control panel mounted).

3.6.2.1 Engine fuel. The diesel engine shall operate efficiently on DG-1, DG-2, DF-A, and DF-M as primary fuels, and be capable of operation on jet fuels JP-4, JP-5, and JP-8.

3.6.2.2 Fuel tank. The fuel tank shall be of sufficient capacity to permit full load operation for a minimum of 18 hours. The fuel tank shall be removable from the floodlight set to permit cleaning and shall contain a fuel shut-off valve at the outlet connection. A fuel tank drain, located on the tank to provide complete emptying of the tank, shall be provided.

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3.6.3 Wiring. Conductors for interconnecting wiring shall be composed of annealed copper wire and have a flame-retardant and oil-resistant covering conforming to NFPA 70. Extra flexible conductors shall be used where relative movement between parts may occur. Wire connectors and terminals shall be the crimp type where practicable and shall be attached in a manner to fully protect the insulation; all wire connectors shall be copper and conform to UL 486. Conductors shall be properly clamped in a neat manner and protected by grommets or other suitable protection from abrasion where the conductors pass through metal.

3.6.3.1 Ballasts. The ballasts for the four 1,000W metal halide lamps shall be the peak-load type designed to operate with 120/240V input. Each ballast shall be the enclosed constant wattage auto-transformer type, and shall be designed for operation at low temperature down to -25 degrees F. The ballasts shall be the plug-in type and mounted near the power distribution panel in a protected location inside the storage compartment. The line side of the ballasts shall be connected to the circuit breakers in the distribution panel.

3.6.3.2 Connectors, plugs, and receptacles. Cord receptacles, cord plugs, and outlet receptacles shall be of the grounding type and shall be in accordance with UL 498, NEMA WD 1 and NEMA WD 5, as applicable. The cord receptacles and cord plugs shall employ cord grip construction and shall be weatherproof. Cord plugs and cord connectors utilizing weatherproof cord grips shall be so designed that they will form a weatherproof seal when the plug is inserted into the mating connector either of another cord or an outlet on the control panel. All receptacles exposed to the weather shall be equipped with spring loaded cover plates.

3.6.3.2.1 Receptacles. Two grounding type two-pole, three-wire, 125V, 15A, general purpose duplex receptacles conforming to NEMA WD 1 configuration 5-15R; and two grounding type three-pole, four-wire, 240V, 20A, single locking type receptacles conforming to UL 498 and NEMA WD 5 configuration L16-20R shall be installed in the same vicinity of each other in an accessible, protected location on the exterior of the storage compartment. The locking type receptacles shall be used for the 500W and 1,000W luminaries. The general purpose duplex receptacles shall be used for auxiliary equipment. Each outlet shall be marked with voltages and current rating. Each duplex receptacle shall be connected to only one circuit so that one circuit breaker protects one duplex receptacle load. The input to each pair of ballasts shall be connected to a multi-wire 120/240V circuit with a common neutral so that two circuit breakers, one for each ungrounded conductor, protect the ballasts. The circuit breaker for each luminary shall also function as the switch for controlling each flood lamp.

3.6.3.2.2 Portable power cord. One portable power cord shall be provided for use as power extension cord for the 500W luminaries mounted on the portable mast. The cord shall total not less than 150 feet in length and shall be 3 conductor, No. 14 American Wire Gage (AWG) copper conductors. One end of the cord shall be terminated in a plug connector conforming NEMA WD 5, configuration L5-15P. The other end of the cord shall be terminated in a receptacle connector conforming to NEMA WD 5, configuration L5-15R. Both plug and connector shall be of weatherproof construction.

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3.7 Storage compartment (housing). A durable metal housing shall be mounted on the trailer to house all permanently mounted power distribution equipment and provide for storage of all floodlights, electrical cords, etc. The housing shall contain hinged doors to allow easy access to stored components and allow access to all components necessary to utilize the floodlight set. The doors shall allow easy access to components requiring servicing. Doors shall be reinforced to provide adequate rigidity and have fasteners to hold them securely in the closed (locked) position. All latches and handles shall be proportioned to facilitate manipulation with heavy mittened hands and with a minimum of projection beyond the housing surface. Doors and hinged covers shall have a lock-open device to securely hold the door or cover in the open position with the generator set in operation. Drip channels or similar features shall be provided over vertically hinged doors and covers. Doors and covers shall be joined with the housing in a manner that will meet environmental requirements and reduce noise resulting from vibration. Means shall be provided for drainage of water entrapped within the storage compartment resulting from operation with doors or covers open. Hinges shall be of the continuous type. The storage compartment may be used for storage of the telescoping mast and the portable mast, provided they are adequately reinforced to provide structural support for extended off-road usage.

3.7.1 Compartment interior lighting. A small light shall be installed to illuminate the interior of the storage compartment. The light shall accommodate a 12V, direct current, incandescent lamp not larger than 40W. The light shall have a frosted dome lens, shall be controlled by a conveniently located toggle switch, and shall be protected by a fuse located at the attaching point to the power source.

3.7.2 Structural support. The compartment shall be reinforced as required to mount and support the lighting fixtures, extension cords and related gear when stowed in the travel positions. The storage compartment shall not permanently deform, twist, or deflect during tests required in 4.6.4.

3.8 Trailer. The trailer chassis shall support the complete floodlight set, when subjected to the usage specified herein, without permanent distortion or damage to any part. The trailer shall be provided with a frame of all steel welded construction. The drawbar shall be of the A-frame type and be an extension of the main side frame members, and may be designed as a quick detachable type to assist the reduction of storage area required. The frame and drawbar shall have structural capacity and be reinforced or braced, where necessary, to produce adequate strength and rigidity to prevent both horizontal and vertical buckling or distortion. The floodlight set shall have a minimum road clearance of 8 inches. Springs shall be of a capacity to support the imposed load when measured at the ground. A lunette ring, typically fabricated from nominal 1-5/8-inch steel rod, with a nominal 3-inch diameter, shall be installed and positioned for attachment to the towing vehicle in such a manner that the coupling height will be adjustable between 20 to 30 inches above the ground.

3.8.1 Landing gear. An adjustable length type landing leg, with a swivel caster wheel, of rugged construction and with provisions to hinge and lock in a vertical and horizontal position, shall be provided and attached to the draw bar. The landing leg shall withstand, without deformation, the combined static and dynamic forces due to proportion of gross weight sustained and the forces

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resulting from impacts due to coupling, decoupling, transportation, and rough handling. The landing leg, when in the vertical position, shall hold the trailer chassis relatively level and shall not obstruct towing in the horizontal position.

3.8.2 Stabilization jacks. Adjustable swivel type jack stands shall be securely attached to extendable and retractable outriggers to level and stabilize the complete, free-standing floodlight set on a 5-degree slope, when tested in accordance with 4.6.5. Means shall be provided for locking the jacks and outriggers in a retracted position which will not obstruct towing. When locking pins are used, they shall be of the captive type with chain or rope. Level indicators shall be provided in accessible protected locations on the trailer chassis to facilitate leveling the floodlight set prior to starting operation. The jacks shall be equipped with swivel foot pads. The stabilization jacks shall provide sufficient support to prevent damage with full extended mast equipped with four floodlights during winds up to 60 miles per hour (mph) continuously from any side direction.

3.8.3 Wheels, rims, and tires. The trailer shall be mounted on two pneumatic-tired wheels manufactured with a highway type tread design and a minimum rim diameter of 13 inches. Wheels shall be demountable at the hub. Rim and tire ratings shall conform to TRA recommendations for the type and size of tires furnished. Tires shall be of the tubeless type. Tires shall be new, unblemished, of first quality, and shall contain anti-ozone compound. Recommended tire pressure shall be stenciled on the trailer near each tire location. Wheel hubs shall have ball or tapered roller bearings.

3.8.4 Fenders. The trailer shall be furnished with fenders to provide protection from moisture or debris being splashed or thrown upon the outer surface of the storage compartment and trailer chassis. When specified (see 6.2), steel fenders shall be provided. Steel fenders shall be constructed and installed in such a manner to serve as a footstep (capable of supporting a 150 lb load) as well as to provide protection from moisture or debris being splashed or thrown upon the outer surface of the storage compartment and trailer chassis.

3.8.5 Center of gravity. The center of gravity of the floodlight set shall be forward of the axle and with all components stowed for towing. The load on the lunette ring shall not exceed 200 lb or be not less than 100 lb.

3.8.6 Tracking ability. With the floodlight set in the stowed position and ready for towing, the floodlight set shall be towed over improved and unimproved road conditions. The floodlight set, when towed in a straight line on an improved road, shall follow the towing vehicle without wandering over 3 inches to either side of the path of the towing vehicle, and shall not induce loads of a magnitude or nature that would cause noticeable steering difficulties to the towing vehicle at towing speeds up to 50 mph. When towed on an unimproved road at speeds up to 15 mph, any resulting forces produced shall not cause any damage to the floodlight set.

3.8.7 Trailer lights, wiring, and reflectors. All trailer lights and reflectors shall conform to DoT Federal Motor Carrier Safety Regulations sections 393.14, 393.19, 393.20, 393.22 through 393.26(d). A lighting cable

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connector conforming to SAE J560 shall be attached to the wiring system to permit connection to the towing vehicle. All wiring in the trailer lighting system shall conform to SAE J1292.

3.8.8 Parking brake. Unless otherwise specified (6.2), a lever type handbrake shall be provided which will hold the floodlight set in a stationary position when tested in accordance with 4.6.10. The brake system shall be an automotive internal expanding shoe and drum type. Parking brake cables shall be made of stainless steel and firmly secured.

3.8.9 Grounding rod and wire. A copper-plated grounding rod of not less than 4 feet in length and not less than 1/2-inch in diameter shall be provided. A grounding wire of not less than 6 feet in length shall connect the grounding rod to the trailer frame. The grounding rod and trailer frame shall each provide a means for attaching the grounding wire. Provisions shall be provided on the trailer for stowing the grounding rod and wire.

3.9 Telescoping tower. The telescoping tower shall be mechanical elevating and telescoping (see 3.9.1). The tower assembly shall have provisions for securely mounting four 1,000W size luminaries in a horizontal configuration with a spacing between luminaries (center-to-center) of not less than 3 feet on the top of the tower when in operating position. The tower shall have four grounding and locking type receptacles mounted in one weatherproof box near the upper end of the top section of the tower. The box shall have a weatherproof cover plate. The box shall supply power to the 1,000W luminaries.

3.9.1 Mechanical telescoping tower. The telescoping tower shall be a fabricated assembly that, when in the upright (operating) position, is continuously adjustable from a fully lowered position of not more than 13 feet to a height of not less than 30 feet above ground. The lower portion of the tower shall be enclosed in a manner to protect operating personnel from injury while the tower is being extended, retracted, or rotated vertically up to 360 degrees. The tower assembly shall be hinged or pivoted in such a manner that, with the tower assembly in the fully retracted position, the assembly may be lowered into a horizontal (transport) position. The floodlight set shall have a positive locking mechanism to secure the tower assembly for all specified operating conditions and secure the tower assembly to the trailer-storage compartment assembly for all specified transport conditions. The floodlight set shall have a manual tower operating mechanism such that one man can raise the tower assembly from the horizontal (transport) vertical position and extend the tower to the maximum height in not more than 5 minutes. The tower assembly shall have a tensioning mechanism to assure that the tower assembly is completely under control of the operating mechanism during extending and retracting operations. The tower assembly shall have structural strength to withstand 60 mph winds from any direction, without damage, when fully extended with the luminaries mounted.

3.9.2 Tower electrical cord. A heavy-duty, flexible, permanently coiled, retractable type power cord shall furnish power to the top of the tower mast. The supply end of the cord shall terminate in the storage housing. The tower mast assembly shall have a guide tube on the side of the mast to encase the power cord when the tower is completely retracted.

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3.9.3 Portable mast. A portable mast assembly with tripod base shall be provided for mounting two 500W size luminaries. The mast assembly shall consist of a manually adjustable telescoping tubular mast with a tubular crossarm on which to mount the luminaries and a collapsible tripod base. The height of the mast shall be manually adjustable, in increments of not more than 6 inches, from 8 feet to a maximum of 12 feet, and shall have a locking mechanism to securely lock the mast at any of the selected positions. The tripod base, when in the operating position, shall securely support the mast in a vertical position at maximum height with the luminaries mounted on the crossarm, and shall provide sufficient stability at this position to withstand 10 mph winds from any direction. The mast, when fully retracted, with luminaries mounted on the crossarm and the base anchored, shall have sufficient structural strength to withstand 30 mph winds from any direction. The portable mast assembly shall be readily collapsible for transport storage in storage compartment (see 3.7). The collapsed mast assembly shall require minimum storage space with an overall length of not more than 6 feet.

3.10 Floodlights. The floodlight set shall have four 1,000W size floodlights and two 500W size floodlights. Each floodlight shall consist essentially of a cast, pressed or spun aluminum, weatherproof housing, with a heat-resistant clear glass lens, and weather-resistant seal. The securing devices shall be captive with either the housing or door. Provisions shall be made to release the securing devices and remove the lamp without tools. The securing devices shall maintain the weather-resistant seal under the wide temperature changes, adverse weather, shock, and vibrational stresses encountered under rough service conditions.

3.10.1 1,000 watt floodlights. The floodlight mounting arrangement shall be such that the four lights attached to the tower may be individually rotated through an arc of not less than 350 degrees in the horizontal plane, and an arc of not less than 135 degrees in the vertical plane. The housing shall have a suitable carrying handle. The floodlight shall have a lateral and vertical beam spread of not less than 120 degrees $\pm$10 degrees. The clear glass lens shall produce a controlled circular beam spread. The floodlight shall be provided with a keyless mogul base lampholder to accommodate a metal halide lamp. The floodlight shall have a shock mount lamp support, and a heat-tempered clear glass lens. Adequate provisions shall be available for front or rear relamping. Each floodlight shall have a flexible power supply cord attached with a watertight cord grip at the floodlight end, and locking type cord plug with weatherproof cover on the supply end. The plug shall conform to UL 498 and NEMA WD 5 configuration, L8-20P. The cord shall be 3 conductor, No. 14 AWG copper, and the green grounding conductor at the floodlight end shall be connected to the floodlight housing. Metal halide lamps which produce white light shall be used. Initial light output for each 1,000W lamp shall be not less than 100,000 lumens.

3.10.2 500 watt floodlight. The two 500W floodlights shall be designed to be mounted on the portable mast. A mounting bracket, adequate to support the floodlight on the portable mast, shall be provided. With the portable mast resting on a horizontal plane, vertical adjustment, without tools, of the floodlight through a range of not less than 60 degrees above to 30 degrees below the horizontal shall be provided. There shall be a means to securely lock the floodlight in the mounting bracket at any selected position. Horizontal adjustment, without tools, of the mounting bracket with respect to the portable

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mast through a full 360 degrees shall be provided. There shall be a means to securely lock the mounting bracket to the portable mast at any selected position. The floodlight shall be designed to use 120V, clear, T-3 rough service, quartz, tungsten halogen lamps with recessed single contact base. The initial light output for 500W lamps shall be not less than 10,900 lumens. A flexible power supply cord shall connect each floodlight to the single-gang weatherproof aluminum box. Each cord shall have a watertight cord grip on each end. The cord shall be 3 conductor No. 16 AWG copper wire. The green grounding conductor at the floodlight end shall be connected to the floodlight housing. A 25-foot long power supply cord shall be attached to the single-gang weatherproof box. The end of the cord attached to the box shall have a watertight cord grip and the other end shall have a grounding type weatherproof plug. The plug shall conform to NEMA WD 5, configuration L5-15P.

3.10.3 Lamps. All lamps shall be designed to operate in a horizontal position.

3.11 Air transportability. When specified (see 6.2), the floodlight set shall be designed to be air transportable in accordance with MIL-STD-1791. Tiedown points, fore, aft, and sidewards shall be marked on the set and shall include the g loads for which each point is designed.

3.12 Lifting and tiedown attachments. When specified (see 6.2), the floodlight set shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to MIL-STD-209, type III. Attachments shall be indicated by a plate conforming to the vehicle shipping data plate described in MIL-STD-209, except tiedown attachments may be identified by stenciling or other suitable marking on the floodlight set. Tiedown marking shall clearly indicate that the attachments are intended for tiedown of the floodlight set on the carrier when shipped.

3.13 Lubrication. Lubrication shall be in accordance with the equipment manufacturer's recommendations.

3.14 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.15 Instruction plates. The floodlight set shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. Plates shall be securely affixed to the equipment with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.15.1 Operating instructions. Plastic encapsulated operating instructions shall be mounted in a convenient place inside the storage compartment. The operating instructions shall be complete and brief.

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3.15.2 Schematic and wiring diagrams. A plastic encapsulated schematic wiring diagram shall be mounted adjacent to the operating instructions. The schematic diagram shall show the complete operation and functional sequence of the circuit for analysis and maintenance of the floodlight set. The wiring diagram shall show the physical location of all wiring interconnections in the same relationship in which they are installed. Identification marking of control devices and connections shall be shown in both diagrams and shall coincide with markings on all items being identified.

3.16 Rustproofing. When specified (see 6.2), rustproofing shall be provided in accordance with MIL-STD-1223.

3.17 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products, or any other interfering substances. As soon as practicable after cleaning and before any corrosion product or other interfering material can result, the surfaces shall be prepared or treated to ensure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat of acrylic-based enamel. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes that produce a total dry film thickness of not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects. The exterior color shall be as specified (see 6.2). The compartment housing interior color may be the manufacturer's standard.

3.18 Workmanship.

3.18.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.18.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.18.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member. Rivets shall be of a nonferrous material.

3.18.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by

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the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.19 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. The first article inspection shall be performed on one floodlight set (see 3.2, 6.2, and 6.4). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.4 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.5, the tests of 4.6.12, and the packaging inspection of 4.7. This inspection shall be performed on the samples selected in accordance with 4.3.

4.5 Examination. Each floodlight shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations

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and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.6 Tests. Unless otherwise specified (see 6.2), the first article floodlight set shall be tested as specified herein. Failure of the floodlight set to meet requirements of the tests herein, even though not performing a test directly related to the specific requirement, shall constitute cause for rejection. The contractor is responsible for providing certification or evidence of conformance to all the requirements specified.

4.6.1 Engine and components test. The engine and components shall be inspected for conformance to requirements of 3.6.1 through 3.6.2.2.

4.6.2 Alternator test. The alternator shall be inspected for conformance to 3.4.

4.6.3 Power distribution equipment test. All electrical components shall be inspected for conformance to 3.6.3 through 3.6.3.2.2. After verification of conformance the floodlight set shall be assembled with all lights connected to the proper outlets. The floodlight set shall be cycled 250 times in 25 separate 10-cycle runs without failure or malfunctioning of any component. A 10-cycle run is as follows:

a. Start engine and turn on the four tower floodlights and the two portable mast floodlights, 10 times, each cycle.
b. The telescoping tower shall be raised to maximum height and lowered to minimum height, one time, each cycle.
c. Turn off lights and shut down engine.
d. Leave engine shut down for 5 minutes.

Note: A minimum of 10 cycles each with engine operating on DF-2 and JP-4 fuels shall be performed during the test operation of 4.6.3.

4.6.4 Storage compartment test. The compartment shall be inspected for conformance with requirements of 3.7.2. The floodlights and all specified and required gear shall be placed in the designated areas and inspected for adequate fit and support. The interior lamp shall be tested for function. Upon completion of the road tests (see 4.6.9) the compartment shall be inspected for cracking, deforming, deflection, twisting and malfunction of the compartment doors and latches. Any sign of distortion or damage shall be cause for rejection.

4.6.5 Stabilization jack test. With the fully elevated floodlight set uncoupled and engine operating and the outriggers extended, the jacks shall prevent any movement of the floodlight set. The fully erected and uncoupled floodlight set shall be subjected to a stability test in accordance with the requirements of 3.8.2.

4.6.6 Telescoping tower tests. With the floodlight set free-standing on a relatively smooth unpaved area having a slope of approximately 5 degrees, the outriggers extended, and the set leveled, the tower shall be raised to its extreme extended position. While extended, a force of 75 lb, to simulate 60 mph wind conditions shall be applied successively to the tower at not less than four

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locations spaced equidistant around its periphery, perpendicular to the tower and at crossarm height, to determine conformance to rigidity and construction requirements.

4.6.7 Center of gravity test. With all components of the floodlight set stowed for normal transporting, the load on the lunette ring shall be measured to determine compliance to 3.8.5.

4.6.8 Fender test. Steel fenders shall be subjected to a static test of a 150-lb load on each fender for a continuous period of 8 hours, or a dynamic load of 75-lb sand bag or similar item attached to the top surface of the fender during the road tests. Any sign of deformation or support failure shall be cause for rejection.

4.6.9 Road test. All components shall be stowed or attached in the normal manner for transporting, with fender weights attached (when applicable) (see 4.6.8), and the floodlight set coupled to a suitable towing vehicle. The floodlight set shall be towed a distance of not less than 50 miles over improved roads (see 6.5.1), at speeds up to and including 50 mph to verify tracking ability conformance to 3.8.6. In addition, the floodlight set shall be towed a distance of not less than 15 miles over unimproved roads (see 6.5.2), at speeds up to and including 15 mph to verify conformance to 3.8.6

4.6.10 Brake holding capacity test. With the complete floodlight set parked on a 15-degree, dry, hard-surfaced incline and with the wheels facing down the slope, the handbrake shall be set and the trailer uncoupled from the towing vehicle and rested on the landing leg to determine conformance to 3.8.8.

4.6.11 Portable mast stability test. The portable telescoping mast shall be fully extended and be subjected to the stability requirements specified in 3.9.3.

4.6.12 Quality conformance test. Each production floodlight set shall be given an operational test. All components and devices shall be operated to determine that they are functioning properly. The electrical system shall be tested for grounds, open circuits, and short circuits. Any set which fails to pass any of the tests shall be rejected.

4.7 Packaging inspection. The preservation, packing, and marking of the floodlight set shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116, and shall conform to the requirements of MIL-P-116 and any applicable specifications.

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5.1.1.2 Cleaning and drying. Prior to the application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Disassembly. Disassembly shall be the minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved with P-1.

5.1.1.5 Generator set. The generator set shall be preserved method I in accordance with the level A requirements of MIL-G-28554 for generator set uncrated.

5.1.1.6 Floodlights, tower, and masts. The floodlights shall be stored in the storage compartment and secured to prevent damage during transit. The telescoping tower shall be retracted to the shipping position and secured. Portable mast shall be collapsed and secured in the storage compartment.

5.1.1.7 Power cords and spare lamps. The power cord and any spare lamps shall be secured in the storage compartment to prevent damage during transit.

5.1.1.8 Jacks. The shaft of the jack shall be extended and preserved with type P-11 preservative. The shaft shall then be fully retracted.

5.1.1.9 Trailer. The trailer shall be preserved in accordance with the level A (mobile) requirements of MIL-V-62038.

5.1.1.10 Lubrication. Lubricate all grease fittings with type P-11 preservation. Wheel bearings requiring a NLGI No. 2 grade grease shall be lubricated with type P-11 preservative. Lubricate control mechanisms, linkage, hinges, fasteners, and all pivot points with P-10.

5.1.1.11 Rubber tires. Tires mounted on the wheels (road tires) of the vehicle shall be inflated to 10 lb above pressure value recommended for maximum load. Tires mounted on rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

5.1.1.12 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.13 Technical publications. Technical publications for each piece of equipment shall be preserved method IC-1 or IC-3.

5.1.1.14 Consolidation. Loose components, tools, and publications for each floodlight set shall be consolidated in containers conforming to PPP-B-636, class weather-resistant. Contents shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186.

5.1.2 Level C. The floodlight set shall be preserved in accordance with the contractor's standard practice in a manner to prevent corrosion, deterioration,

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and damage. The floodlight set shall be lubricated and serviced for operation as required in the operator's manual and as specified in section 3.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Crated floodlight sets. Unless otherwise specified (see 5.2.1.2 and 6.2), each floodlight set shall be packed in a crate conforming to MIL-C-104, type II, class 1 or 2, style A. Anchoring, cushioning, blocking, and bracing of the contents shall be in accordance with the appendix to MIL-C-104. Strapping shall conform to ASTM D 3953. The floodlight set shall not be crated resting on its supporting legs (landing gear).

5.2.1.2 Mobile (uncrated) floodlight sets. When specified (see 6.2), each floodlight set shall be prepared for mobile shipment in accordance with applicable carrier rules and regulations. Consolidation containers shall be secured on the floodlight set in a manner not to increase cubage or interfere with towing or lifting the floodlight set with slings. Do not ship the floodlight set resting on its support legs (landing gear).

5.2.2 Level B.

5.2.2.1 Crated floodlight sets. Unless otherwise specified (see 5.2.2.2 and 6.2), each floodlight set shall be packed in a crate conforming to MIL-C-3774, type II, style A. Anchoring, cushioning, blocking, and bracing of the contents shall be in accordance with the appendix to MIL-C-3774. The floodlight set shall not be crated resting on its stabilization jacks or landing gear.

5.2.2.2 Mobile (uncrated) floodlight sets. When specified (see 6.2), each floodlight set shall be prepared for mobile shipment as specified for level A.

5.2.2.3 Level C. The equipment shall be prepared for shipment in a manner which will insure arrival at destination in a satisfactory condition. Preparation for delivery shall comply with applicable carrier rules and regulations. Do not ship the floodlight set resting on its support legs (landing gear).

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The floodlight set is intended for use at shore establishments and for military field application under all kinds of climatic conditions. It is a completely self-contained unit, suitable for use as a source of electrical floodlighting in remote locations, with electrical power as an auxiliary feature.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. First article tests required (see 3.2, 4.3, 4.6, and 6.4).
d. When steel fenders shall be provided (see 3.8.4).
e. When a parking brake is not required (see 3.8.8).
f. When air transportability is required (see 3.11).
g. When lifting and tiedown attachments are required (see 3.12).
h. When fungus resistance treatment is required (see 3.14).
i. When rustproofing is required (see 3.16).
j. Exterior color required (see 3.17).
k. Level of preservation and level of packing required (see 5.1 and 5.2).
l. Crating requirements (see 5.2.1.1 through 5.2.2.2).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL, DD Form 1423), incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations (FAR) Supplement, Part 27, Sub-Part 27.475-1 are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required, the floodlight will be tested and approved under the appropriate provisions of supplement 27.410-5 of the FAR and should be a first production item as specified in 4.3. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitions.

6.5.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete or asphalt paved highway.

6.5.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

6.5.3 Recovered materials. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials.

6.6 Certifications. Certificates of compliance from the manufacturer should be considered in lieu of testing for:

a. Manufacturer's certification that the operating components furnished meet all the conditions stated in paragraph 3.1.2, subparagraphs a and b may be acceptable.

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b. Acceptable evidence of meeting the applicable UL standards cited in this specification may be the UL listing mark affixed to the item certified, or a certified test report from a nationally recognized independent testing laboratory, acceptable to the contracting officer, stating that the item certified has been tested and conforms to the applicable UL standards.

6.7 Subject term (key word) listing.

Portable light unit
Power supply unit
Powered light unit

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Navy - YD1

Review Activity:
DLA - GS

User Activity:
Navy - MC

Preparing Activity:
Navy - YD1

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2. 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-29161D	2. DOCUMENT DATE (YYMMDD) 930820

3. DOCUMENT TITLE
FLOODLIGHT SET, ELECTRIC, DIESEL-ENGINE-DRIVEN, TRAILER MOUNTED

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
RUSSELL B. REYNOLDS	805-982-5946	551-5946
c. ADDRESS (Include Zip Code) COMMANDING OFFICER, NCBC CODE 1562C 621 PLEASANT VALLEY ROAD	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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